Detection of PWM Signal:

- The working of the PWM detector can be explained with following block diagram (Fig.1) and waveforms are shown in Fig. 2.

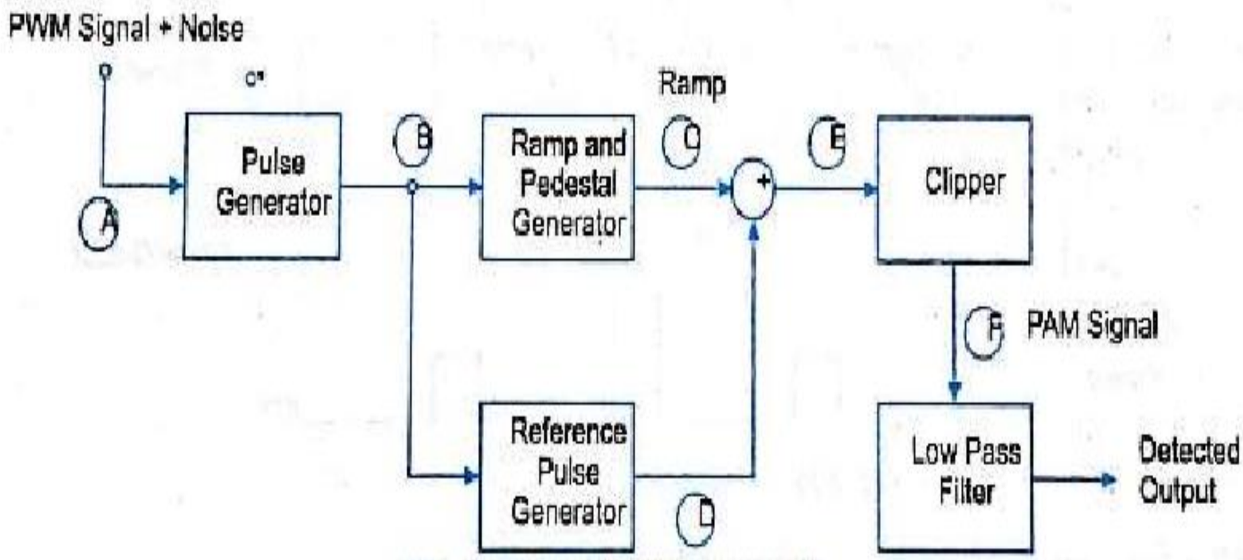

Fig. 1 Block diagram of PWM detection

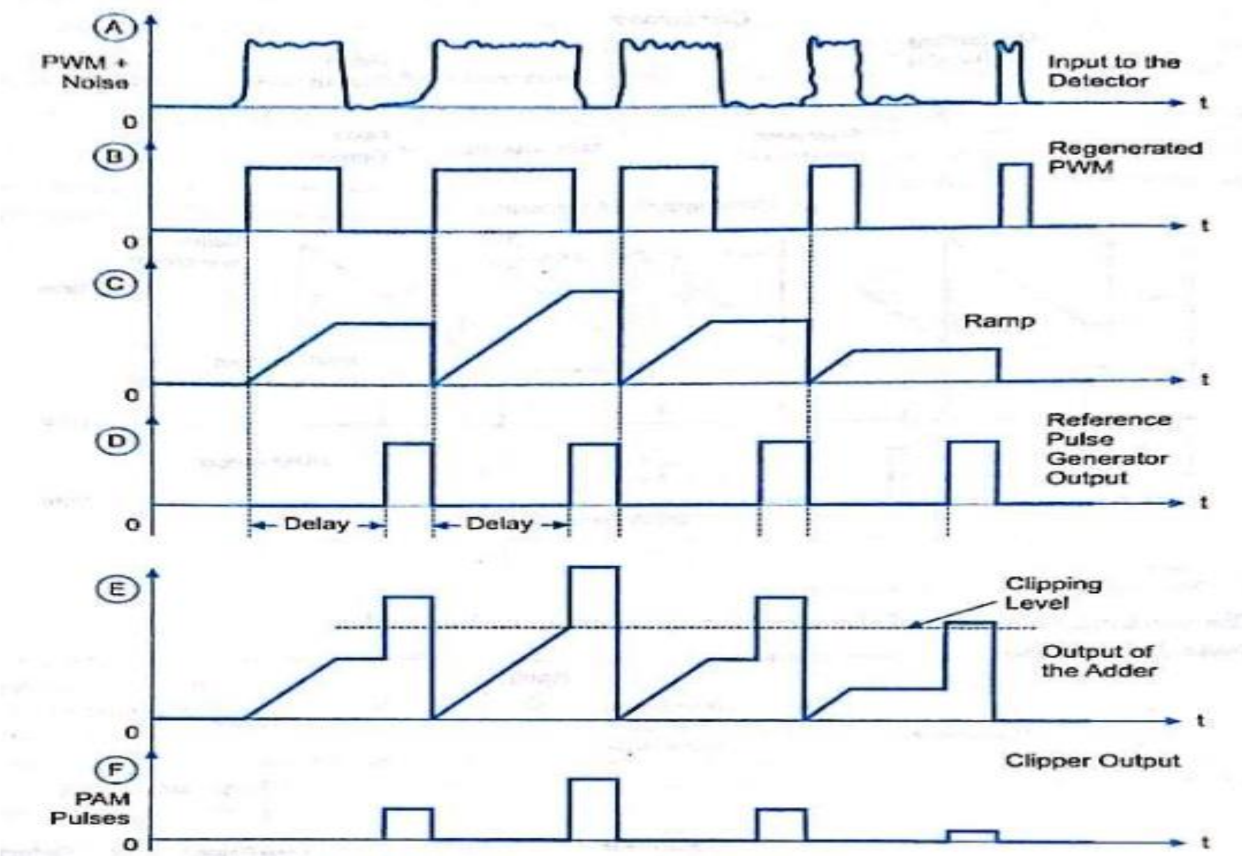

Fig. 2 Waveforms of PWM detection

- The PWM signal received at the input of the detection circuit is contaminated with noise. **(Waveform A)**. This signal is applied to pulse generator circuit which regenerates the PWM signal. **(Waveform B)**
- The regenerated pulses are applied to a reference pulse generator; it produces a train of constant amplitude, constant width pulses. **(Waveform D)**
- These pulses are synchronized to the leading edges of the generated PWM pulses but delayed by a fixed interval. The regenerated PWM pulses are also applied to a ramp generator.
- At the output of it, **(Waveform C)** we get a constant slop ramp for the duration of the pulse. The height of the ramp is thus proportional to the width of the PWM pulses.
- At the end of the pulse, a sample and hold circuit retains the final ramp voltage until it resent at the end of the pulse.
- The constant amplitude pulses at the output of reference pulse generator are then added to the ramp signal. **(Waveform E)**
- The output of the adder is then clipped off at a threshold level to generate a PAM signal at the output of the clipper. **(Waveform F)**
- A low pass filter is used to recover the original modulating signal back from the PAM signal.

Detection of PPM Signal:

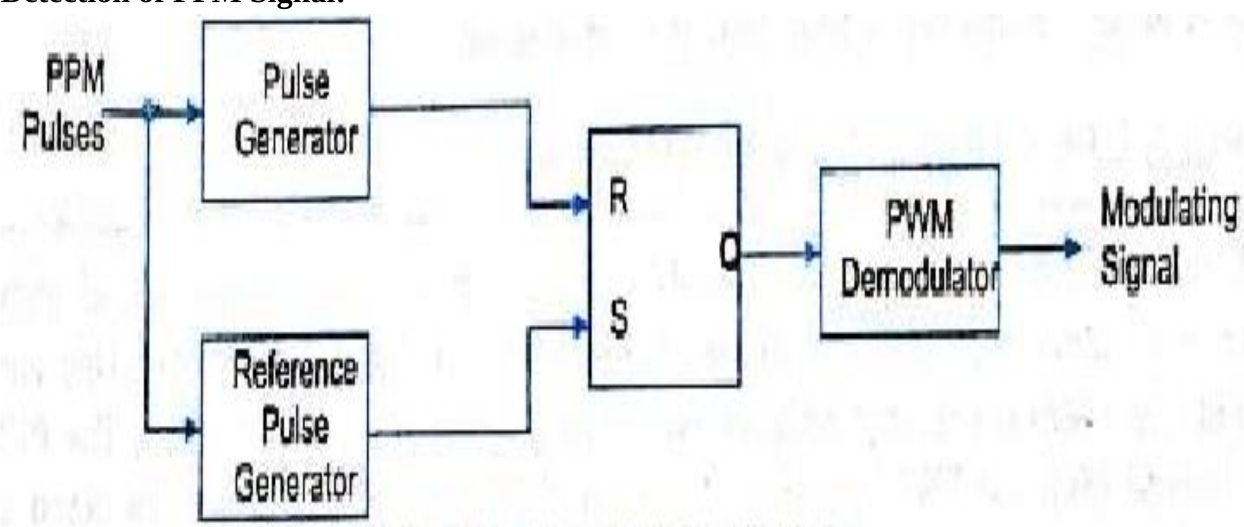

Fig. 3 Block diagram of PPM detection

- The noise corrupted PPM waveform is received by the PPM demodulator circuit. The pulse generator develops a fixed duration pulses at its output and it is applied to reset pin (R) of a SR flip flop.
- A fixed period reference pulse is generated from the incoming PPM waveform and the SR flip flop is set by the reference pulses.
- Due to set and reset signals applied to the flip-flop, we get a PWM signal at its output. The PWM signal can be demodulated using the PWM demodulator.